FTL – Syria Pair coding

Problem 1

Which communities report the greatest water interruptions?

Context: A community survey team has collected information about water availability. Before using the survey, your team must clean the records and identify where further investigation is needed.

Your task

1. Load the survey dataset
2. Inspect its structure, data types, missing values and duplicate records.
3. Apply these rules:
    - Keep one record per survey_id.
    - Convert households and days_without_water to numeric values.
    - Households must be positive whole numbers.
    - Days without water must be whole numbers between 0 and 7.
    - Exclude incomplete or invalid records from the analysis, but retain them in a separate review table.
4. For each community, calculate:
    - Total households represented.
    - Households reporting **at least three days without water**.
    - Percentage of represented households meeting that condition.
5. Produce a bar chart comparing those percentages.
6. Recommend which community to investigate first, explaining your reasoning.

Data collection question: Write two validation rules you would add to the survey form to prevent these errors.

Useful tools: pd.read_csv(), pd.to_numeric(), duplicated(), drop_duplicates(), filtering, groupby() and plotting.

Problem 2

Where are water interruptions concentrated?

Context: A programme coordinator wants a map of surveyed communities to help plan follow-up visits.

Your task

1. Clean the same survey dataset using the rules from Problem 1.

2. Aggregate the survey to **one row per community**.
3. Calculate the percentage of represented households reporting at least three days without water.
4. Check that each community has consistent coordinates. Flag conflicting coordinates rather than averaging them automatically.
5. Create a GeoDataFrame:
    - Longitude is **x**.
    - Latitude is **y**.
    - Use **EPSG:4326** for the supplied coordinates.
6. Plot a point map:
    - Colour represents the percentage affected.
    - Label each community.
    - Include a title and legend.
7. Recommend two communities for follow-up visits.

Data collection question: What location information should the survey team record, and how could they check its accuracy?

Deliverables: A community GeoDataFrame, one map and a short explanation of the two recommended visits.

Useful tools: groupby(), nunique(), gpd.GeoDataFrame(), gpd.points_from_xy() and .plot().

Important: No online basemap is required. Do not calculate distances directly from latitude and longitude in degrees.

Optional extension: Compare priorities based on the **percentage affected** versus the **number of households affected**. Does the recommendation change?

Problem 3

Scenario: Your team is preparing a climate-awareness presentation. Use daily temperature data for a selected point near Aleppo to identify when hot days occurred during 2024.

For this exercise, define a **hot day as a day with maximum temperature of at least 35°C**. This is a classroom threshold.

Dataset and links

Source: **NASA POWER**, parameter T2M_MAX—daily maximum temperature. These are gridded estimates for the selected location, rather than measurements from a local weather station.

- NASA POWER documentation

Student tasks

1. Load the data and inspect the first five rows.
2. Check dates, missing temperatures and duplicate dates.
3. Count valid temperature observations in each month.
4. Create a column identifying hot days.
5. Calculate the number of hot days per month.
6. Create a bar chart with month labels and a clear title.
7. Answer:
    - Which month had the most hot days? Report ties.
    - What was the highest daily maximum temperature, and on which date or dates?
    - What practical activity could a community organisation consider during the hottest period?

Missing-data rule: An unknown temperature must not be counted as a cool day. Report missing observations alongside the monthly results.